AI Assistance & Attribution Disclosure

Quantum Traffic Gridlock — IBM Qiskit Fall Fest 2026 (QC@IITI, IIT Indore)

In compliance with Section 10 (*"Mathematical Integrity & AI Use"*) of the challenge specification, this document transparently describes the use of AI assistance, libraries, and independent verification throughout the development of this project.

1. Scope and Nature of AI Assistance

An agentic AI assistant was used as a pair-programming partner to assist with:

- **Code Structuring & Implementation:** Generating initial scaffolding for the modular project architecture (`scenario.py`, `traffic_model.py`, `qubo_builder.py`, `ising.py`, `qaoa_solver.py`, `baselines.py`, `metrics.py`, `serialization.py`, `run.py`).
- **Mathematical Algebra Double-Checking:** Verifying the analytical expansion of quadratic forms in the QUBO-to-Ising conversion matrix elements and confirming tensor sign conventions.
- **Unit Testing Scaffolding:** Drafting automated pytest test cases for edge conditions (zero arrivals, storage overflows, transfer rounding).

2. Independent Human Verification & Scientific Derivation

All critical scientific and mathematical components were independently validated:

1. **Traffic Dynamics Verification:** The four published reference vectors (`1010|1010 → 116.0`, `1111|1111 → 146.0`, `0000|0000 → 237.0`, `0111|1001 → 92.5`) were independently computed and unit-tested to guarantee zero deviation from official scoring dynamics.
2. **First-Principles QUBO Derivation:** No answers, optimal schedules, or scenario-specific parameters were hardcoded or back-propagated into the QUBO matrix. The quadratic model was derived from fundamental physical traffic equations (effective queue departure constraints and marginal overflow penalty scaling).
3. **Algebraic Invariance:** Exact state-by-state numerical equivalence between the QUBO matrix and the Ising Hamiltonian was verified across all 256 computational basis states, ensuring machine-precision agreement ($\leq 1.71 \times 10^{-13}$).
4. **Quantum Sampling Authenticity:** All 4,096 measurement counts reported in `counts.json` were genuinely sampled from the final bound parameterized circuit executed on Qiskit's `AerSimulator`. No counts or benchmark results were synthesized, fabricated, or post-processed.
5. **Separation of Hybrid Methods:** Classical local refinement was kept strictly separate from the raw quantum sampling metrics.

3. Libraries & Environment

Component	Version
Python	3.12.5
Quantum Computing Framework	Qiskit 2.5.2, Qiskit Aer 0.17.2
Scientific Computing	NumPy 2.1.1, SciPy 1.16.1
Testing	Pytest 9.1.1